

Bounded-Segment Crossover: M3 Model Verification

Summary of `crossover.py`

(a) Theory

The M3 model

The M3 model describes a polymer chain of length L whose tube is decomposed into at most N_e segments, each of length at least s . The accessible configurations are counted by

$$P_L^{(N_e, s)} = \#\{\text{partitions of } L \text{ into at most } N_e \text{ parts, each } \geq s\}$$

This is a doubly constrained partition problem: the upper bound $k \leq N_e$ limits the number of segments (the tube entanglement constraint), and the lower bound $r_i \geq s$ enforces a minimum segment length (the persistence length constraint). For $s = 1$ the lower bound is trivially satisfied, so the constraint reduces to partitions of L into at most N_e parts.

Note: the script's internal documentation refers to this as the "M4 model", using an earlier numbering. In the paper, M3 is the bounded-segment model and M4 is the distinct-parts model.

Two asymptotic regimes

For small L the upper bound $k \leq N_e$ is rarely active and $P_L^{(N_e, s)}$ is essentially the unrestricted partition number $p(L)$. The conformational entropy is therefore governed by the Hardy–Ramanujan formula:

$$\log P_L^{(N_e, s)} \sim \alpha_0 \sqrt{L}, \quad \alpha_0 = \pi \sqrt{\frac{2}{3}} \approx 2.565, \quad L \ll L^*$$

For large L the constraint $k \leq N_e$ binds on every configuration. Using the standard asymptotic $P_L^{(N_e, s)} \sim (L/s)^{N_e-1} / (N_e! (N_e - 1)!)$ for fixed N_e and large L , the entropy grows only polynomially:

$$\log P_L^{(N_e, s)} \sim (N_e - 1) \log(L/s), \quad L \gg L^*$$

Two crossover scales

The script computes two characterisations of the crossover location.

Grand canonical estimate. Setting $E[k]_{\text{GC}} = N_e$ where $E[k]_{\text{GC}} \sim (\pi/\sqrt{6})\sqrt{L}$ gives the closed-form expression:

$$L_{\text{GC}}^* = \frac{6}{\pi^2} N_e^2 \approx 0.608 N_e^2$$

Canonical estimate. Setting the canonical mean $E[k]_{\text{canon}} = N_e$ where $E[k]_{\text{canon}} \sim (\sqrt{6}/2\pi)\sqrt{L} \log L$ gives L_{canon}^* as the solution of:

$$\frac{\sqrt{6}}{2\pi} \sqrt{L^*} \log L^* = N_e$$

solved numerically by bisection. For $N_e = 10, 20, 30$: $L_{\text{canon}}^* \approx 45, 116, 208$ — substantially smaller than the GC values $L_{\text{GC}}^* \approx 61, 243, 547$. The ratio $L_{\text{GC}}^*/L_{\text{canon}}^*$ grows from ≈ 1.3 at $N_e = 10$ to ≈ 2.6 at $N_e = 30$ and ≈ 4.4 at $N_e = 85$, because the logarithmic factor in $E[k]_{\text{canon}}$ pushes the canonical crossover to shorter chains. L_{canon}^* is the physically correct crossover estimate and is what the paper uses throughout.

Asymptote-intersection scale

The script also computes $L^\dagger$, the chain length at which the two asymptotic predictions are equal: $\alpha_0 \sqrt{L} = (N_e - 1) \log(L/s)$. This is not the onset of the crossover but the point at which the polynomial asymptote has caught up with the exponential one. For $N_e = 10, 20$: $L^\dagger \approx 465, 3705$; for $N_e = 30$, $L^\dagger > 5000$.

Physical significance

The crossover at L_{canon}^* marks the onset of strong tube confinement. The N_e^2 scaling of L^* is the robust quantitative prediction, with $L^*/N_e \sim N_e$ corresponding to a chain of approximately N_e entanglement lengths.

(b) What the script does

The script accepts N_e values, minimum segment length s , and L_{max} via command-line arguments. Default values are $N_e \in \{10, 20, 30, 50, 75, 85, 100, 200\}$, $s = 1$, $L_{\text{max}} = 5000$. For each N_e value it computes:

1. The exact M3 partition count $P_L^{(N_e, s)}$ via dynamic programming.
2. The unrestricted $p(L)$ for reference (computed once and shared).
3. A crossover table showing $\log P_L^{(N_e)}$, the exponential asymptote $\alpha_0 \sqrt{L}$, and the polynomial asymptote $(N_e - 1) \log(L/s)$, with both asymptote ratios, at logarithmically spaced L values anchored to L_{GC}^* .
4. A summary crossover table showing L_{GC}^* and $L^\dagger$ for all N_e values.
5. Three CSV files: full crossover data, a crossover summary, and scaled entropy data ($\log P$ and both ratios versus L/L_{GC}^*).

The header also displays both L_{GC}^* and L_{canon}^* for all N_e values before the per- N_e analysis begins.

(c) How the script does it

Unrestricted partitions: `compute_p`

Uses Euler's pentagonal number recurrence (same as in `rr_entropy.py`): $O(L\sqrt{L})$ time, $O(L)$ space. Computed once for all $L \leq L_{\text{max}}$ and reused across all N_e values.

M3 partition count: `compute_m4`

Builds a 2D table `dp[n][k]` = number of partitions of n into exactly k parts each $\geq s$, using the recurrence:

$$p_k^{(\geq s)}(n) = p_{k-1}^{(\geq s)}(n - s) + p_k^{(\geq s)}(n - k)$$

The first term counts partitions whose smallest part equals s (remove it to get a partition of $n - s$ into $k - 1$ parts $\geq s$); the second counts those whose smallest part exceeds s (subtract 1 from every part to get a

partition of $n - k$ into k parts $\geq s$). The table has dimensions $(L_{\max} + 1) \times (N_e + 1)$ and takes $O(L_{\max} \cdot N_e)$ time and space. The M3 count is the row sum: $P_L^{(N_e, s)} = \sum_{k=1}^{N_e} \text{dp}[L][k]$.

Crossover analysis: `analyse_ne`

For each N_e , evaluates $\log P_L^{(N_e)}$, $\alpha_0 \sqrt{L}$, and $(N_e - 1) \log(L/s)$ at a set of L values: $\{0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 1.0, 1.5, 2.0, 3.0, 5.0, 8.0, 12.0, 18.0\} \times L_{\text{GC}}^*$, clipped to $[N_e s, L_{\max}]$. Reports the two asymptote ratios $r_{\text{exp}} = \log P / (\alpha_0 \sqrt{L})$ and $r_{\text{poly}} = \log P / [(N_e - 1) \log(L/s)]$.

The asymptote-intersection scale $L^\dagger$ is found by scanning L from $N_e s$ to $L_{\max}$ and detecting the sign change of $\alpha_0 \sqrt{L} - (N_e - 1) \log(L/s)$.

Canonical crossover scale

Computed in the header block by bisection over $[2, 10^9]$ to 100 iterations, solving $(\sqrt{6}/2\pi) \sqrt{L} \log L = N_e$ to full floating-point precision.

(d) Output produced

Header table (all N_e values)

N_e	L_{GC}^*	L_{canon}^*
10	60.8	45.3
20	243.2	116.3
30	547.1	207.9
50	1520.4	444.0
75	3420.9	827.3
85	4394.8	997.1
100	6079.3	1312.8
200	24317.0	3856.8

Per- N_e crossover tables ($s = 1$, $L_{\max} = 5000$)

For each N_e the table shows L , L/L_{GC}^* , $\log P$, the exponential asymptote, the polynomial asymptote, and both ratios. Key observations from the data:

For $N_e = 10$: at $L/L_{\text{GC}}^* = 1$ ($L = 60$), the ratio $\log P / (\alpha_0 \sqrt{L}) \approx 0.613$, showing the chain is already well into the crossover zone. The polynomial ratio is only 0.33 at $L/L^* = 1$, confirming the transition is gradual.

For $N_e = 20$: the exponential ratio peaks near $L/L_{\text{GC}}^* \approx 0.70$ at ≈ 0.732 , then decreases. At $L/L^* = 1$ ($L = 243$) the ratio is 0.729.

For $N_e = 30$: the exponential ratio peaks around $L/L_{GC}^* \approx 0.70$ at ≈ 0.786 ; the polynomial ratio reaches only 0.41 at the maximum available $L = 5000$.

Crossover summary

N_e	L_{GC}^*	$L^\dagger$ (asymptotes equal)
10	60.8	465
20	243.2	3705
30	547.1	N/A (> 5000)

The ratio $L^\dagger/L_{GC}^*$ grows rapidly with N_e (from 7.6 at $N_e = 10$ to 15.2 at $N_e = 20$), confirming that the polynomial asymptote is not a practical description at accessible chain lengths for $N_e \geq 30$.

CSV files

Three files are written to the same directory as the script (with an optional `--prefix` prepended):

- `_m4_crossover_data.csv`: full per- L data for all N_e values.
- `_m4_crossover_summary.csv`: L_{GC}^* , $L^\dagger$, and their ratio, one row per N_e .
- `_m4_scaled_entropy.csv`: $\log P$, both asymptotes, and both ratios versus L/L_{GC}^* , for use in scaled crossover figures.

(e) What we learn

The crossover is real but very gradual. The entropy ratios $\log P/(\alpha_0 \sqrt{L})$ and $\log P/[(N_e - 1) \log L]$ both remain well below 1 throughout the computed range. Even at $L/L_{GC}^* \approx 20$ for $N_e = 10$, the polynomial ratio is only 0.57. The transition from exponential to polynomial entropy extends over more than a decade in reduced chain length L/L^* , and the asymptotes do not cross until $L^\dagger \sim 8\text{--}15 \times L_{GC}^*$.

L_{canon}^* is the correct crossover onset; L_{GC}^* overestimates it. The factor by which L_{GC}^* overestimates the onset grows from ≈ 1.3 at $N_e = 10$ to ≈ 2.6 at $N_e = 30$ and ≈ 4.4 at $N_e = 85$. For quantitative comparison with simulation data, L_{canon}^* should be used.

The polynomial exponent is $N_e - 1$, not N_e . The count $P_L^{(N_e, s)} \sim (L/s)^{N_e-1}/(N_e! (N_e - 1)!)$ gives entropy $\sim (N_e - 1) \log(L/s)$. The data confirm this: at $L = 1215$ for $N_e = 10$, $\log P = 36.2$ compared to the polynomial prediction $(N_e - 1) \log L \approx 63.9$, with the ratio 0.57 converging from below toward 1.

The N_e^2 scaling of L^* is robust. Both $L_{GC}^* \propto N_e^2$ and L_{canon}^* (which scales approximately as $N_e^2 / \log^2 N_e$) grow quadratically with N_e . This provides a partition-theoretic derivation of the onset of the entangled regime, with $L^*/N_e \sim N_e$ corresponding to a chain of approximately N_e entanglement lengths.

The script covers a broader N_e range than described in the paper. The default run includes $N_e \in \{10, 20, 30, 50, 75, 85, 100, 200\}$; the paper's figures use $N_e \in \{10, 20, 30\}$. The extended range confirms that the qualitative crossover behaviour and the N_e^2 scaling persist across the full range of physically relevant entanglement lengths.